

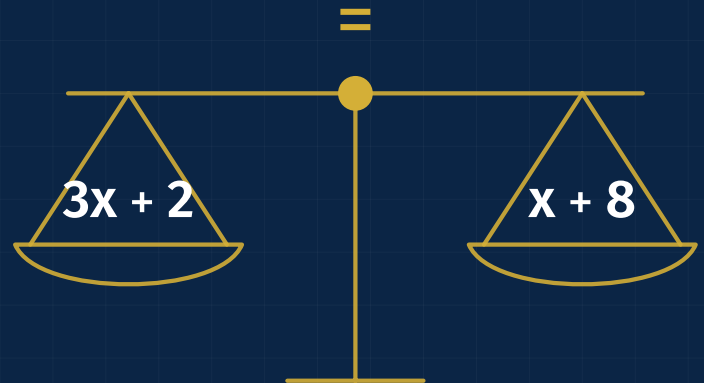


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CHAPTER 2 • ALGEBRA • TOPIC 6

Algebraic Fractions

One golden rule — factorise everything first. Then cancel, KFC for $\div$, the LCM for $+$ and $-$, and $\times$ every term to solve.



7

VARIANTS

18

PAST-PAPER PARTS

52

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Factorise & cancel

V2 $\times$ and $\div$ (KFC)

V3 $+$ $-$ simple bottoms

V4 $+$ $-$ bracket bottoms

V5 Mixed operations

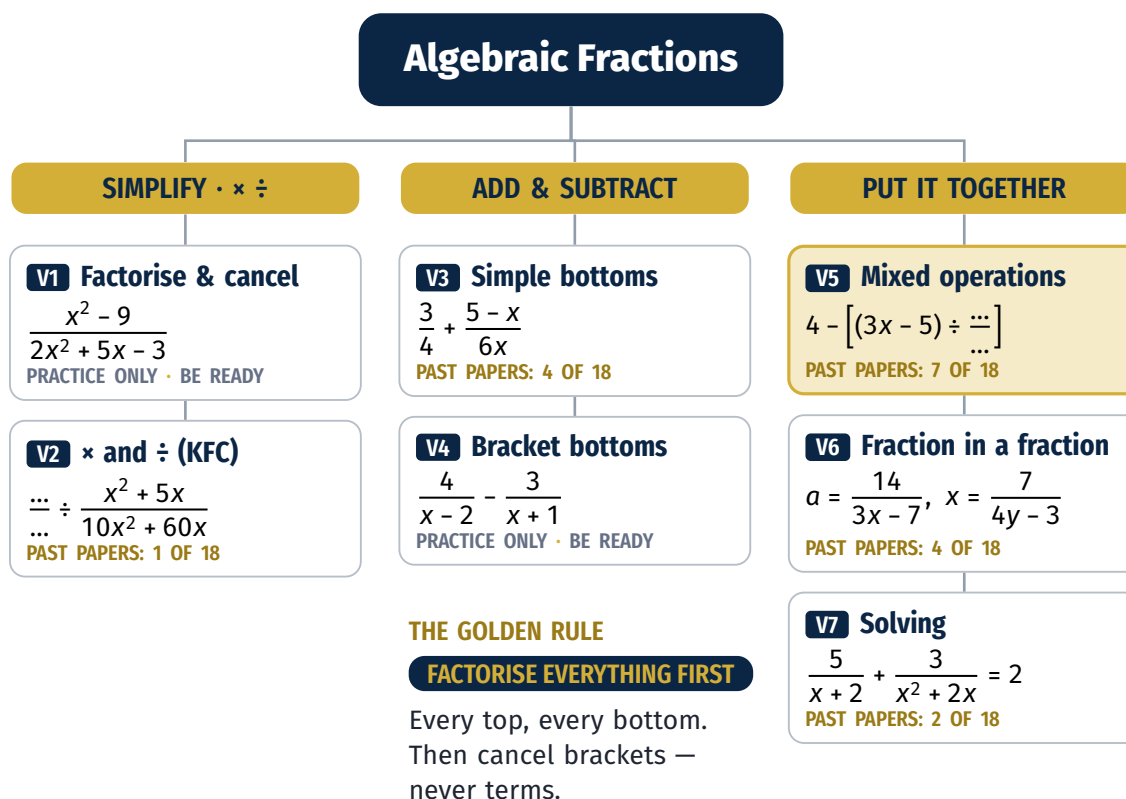
V6 Fraction in a fraction

V7 Solving

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Algebraic Fractions at a glance



Your notes


★ How often does each variant come up?

18 past-paper questions (Jan 2020 → Nov 2025), one fraction part each; 52 practice parts — grouped by type.

	PAST PAPERS (OUT OF 18)	PRACTICE PARTS
V1 Factorise & cancel	0	14
V2 × and ÷ (KFC)	1	3
V3 + - simple bottoms	4	7
V4 + - bracket bottoms	0	12
V5 Mixed operations	7	5
V6 Fraction in a fraction	4	1
V7 Solving	2	10

WHAT THIS TELLS YOU The exam loves **combinations** (V5, V6: 11 of 18) — 3–5 marks near the end of the paper. V1–V4 are the pieces they are built from.

BEFORE YOU START

The big idea



Algebraic fractions follow number-fraction rules.

$\frac{6}{8} = \frac{2 \times 3}{2 \times 4} = \frac{3}{4}$: you cancel a **common factor**. In algebra the factors are **brackets** — so you must **factorise** first (Topic 4) to see them.

$$\frac{6}{8} = \frac{\cancel{2} \times 3}{\cancel{2} \times 4} = \frac{3}{4}$$

$$\frac{x^2 - 9}{x^2 + 3x} = \frac{(x+3)(x-3)}{x(x+3)}$$

Which tool for which sign?

Simplify

$$\frac{(x+3)(x-3)}{x(x+3)}$$

Factorise, cancel the same bracket.

×

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Factorise, cancel across, multiply.

÷ KFC

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Keep · Flip · Change.

+ - LCM

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Same bottom first.

Cancel brackets, never terms

✓ A BRACKET THAT MULTIPLIES EVERYTHING

$$\frac{2(x+3)}{x(x+3)} = \frac{2}{x}$$

× A TERM IN A SUM

$$\frac{x+4}{x+6} \neq \frac{4}{6} \text{ (it is finished)}$$



CALCULATOR FIRST, THEN CHECK

Check any answer with **CALC**: type the question with $x = 2$ (or $x = 5$ if 2 makes a bottom 0), then your answer with the same x . Equal values? You are right. This works for every variant in this topic.

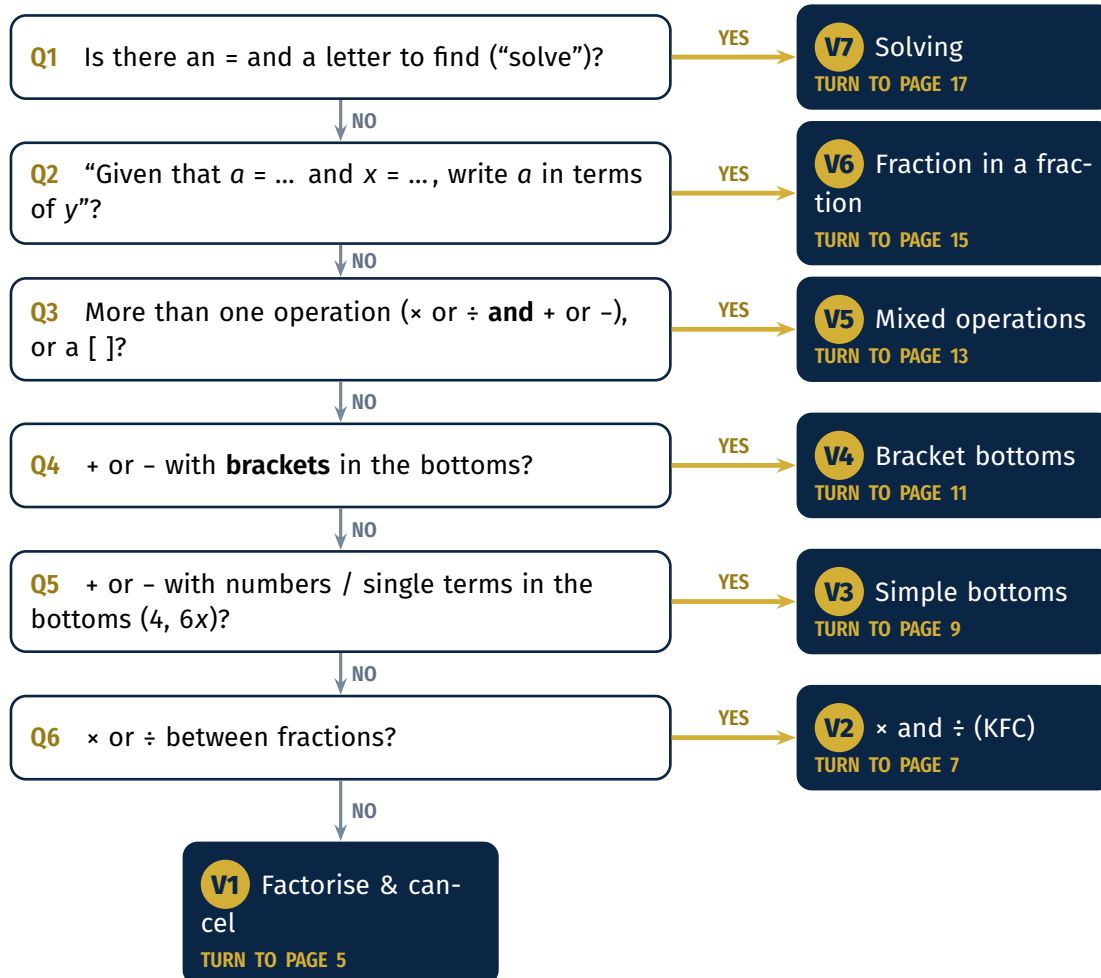
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Simplify: Factorise & Cancel

“Simplify fully” one fraction: factorise top and bottom, cancel the same bracket.

SEEN IN PAST PAPERS

0 / 18

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 14

 Your notes

? What is it?

IN PLAIN WORDS

A fraction with algebra top and bottom. **Factorise both** (HCF, DOTS, CC Rule – Topic 4). A bracket that appears on the top **and** the bottom cancels. If $(3 - x)$ meets $(x - 3)$, take out -1 first.

IT LOOKS LIKE

$$\frac{4(x+5)}{x^2+2x-15}$$

$$\frac{x^2-9}{2x^2+5x-3}$$

$$\frac{8x^2-8}{4x+4}$$

$$\frac{x^2-9}{2x^2+5x-3} \xrightarrow{\text{FACTORISE}} \frac{(x+3)(x-3)}{(2x-1)(x+3)} \xrightarrow{\text{CANCEL}} \frac{x-3}{2x-1}$$

top: DOTS · bottom: CC Rule

Factorise both, then the matching bracket cancels – the answer cannot be simplified any more.

How do I do it?

1 Factorise the top

Count the terms: HCF, DOTS or CC Rule.

2 Factorise the bottom

Same choice. Often one bracket is the same as on top.

3 Cancel

Cross out a bracket that is on the top and the bottom.

4 Check

Nothing else cancels; CALC with $x = 2$.

FACTORISE EVERYTHING FIRST Dr Eslam's golden rule.

V1 Simplify: Factorise & Cancel

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Simplify fully" + one fraction*→ **Factorise top and bottom, cancel***(3 - x) with (x - 3)*→ **3 - x = -(x - 3): cancel, keep a minus***"Show that ... simplifies to ax + b"*→ **Factorise, cancel — the bottom disappears****WORKED EXAMPLE** **PRACTICE**

Algebraic Fractions · Hard Q8 · 3 marks

Simplify $\frac{x^2 - 9}{2x^2 + 5x - 3}$

STEP 1 $x^2 - 9 = (x + 3)(x - 3)$

◀ top: DOTS

STEP 2 $2x^2 + 5x - 3 = (2x - 1)(x + 3)$

◀ bottom: CC Rule (roots $\frac{1}{2}, -3$)

STEP 3 $\frac{(x+3)(x-3)}{(2x-1)(x+3)}$

◀ cancel the same bracket

ANSWER $\frac{x - 3}{2x - 1}$

✓ CALC x = 2: both sides $-\frac{1}{3}$ **WATCH OUT**

✗ $\frac{x-3}{2x-1} = \frac{-3}{2-1}$

✓ $\frac{x-3}{2x-1}$ is finished

You cannot cancel the x's in a sum — only whole brackets that multiply.

EXAM TIP

Expect one bracket to cancel. If none does, re-check your factorising — usually a sign.

Check yourself

calculator first · answers on page 19

1.1 Practice · MediumSimplify $\frac{4(x+5)}{x^2+2x-15}$

Answer:

1.2 Practice · MediumSimplify fully $\frac{10x^2+23x+12}{4x^2-9}$

Answer:

1.3 Practice · HardShow that $\frac{8x^2-8}{4x+4}$ simplifies to $ax + b$.

Answer:

V2

VARIANT 2

**Multiply & Divide
(KFC)**

×: factorise and cancel across. ÷: Keep, Flip, Change first.

SEEN IN PAST PAPERS

1 / 18

PRACTICE QUESTIONS: 3



Your notes

Came up in: Jun 2025 P2H Q23

? What is it?**In plain words**

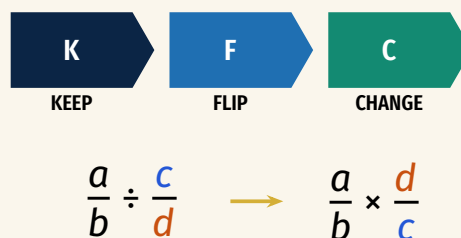
Multiply: factorise every top and bottom, cancel any bracket on a top with the same on a bottom, multiply what is left.

Divide: **KFC** — Keep the first, Flip the second, Change ÷ to ×. Then it is a multiply.

IT LOOKS LIKE

$$\frac{25a}{8} \times \frac{2a}{5}$$

$$\frac{3x+6}{x^2-3x-10} \div \frac{x+5}{x^3-25x}$$

**How do I do it?****1 KFC (if ÷)**

Keep the first fraction, flip the second, ÷ becomes ×.

2 Factorise all

Every top and every bottom — four factorisations.

3 Cancel across

A bracket on any top cancels with the same on any bottom.

4 Multiply

Multiply what is left: tops together, bottoms together.

KFC: KEEP • FLIP • CHANGE

Flip only the fraction after the ÷.

V2 Multiply & Divide (KFC)

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO $\div$ between two fractions→ **KFC, then factorise and cancel**“= p where p is an integer”→ **Everything cancels to a number**“Show that ... simplifies to ax ”→ **Factorise fully (HCF, then DOTS) and cancel****WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 2H · Q23 · 4 marks

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p \text{ where } p \text{ is an integer. Find } p.$$

STEP 1	$\frac{4x^2 - 4x - 120}{5x^2 - 180} \times \frac{10x^2 + 60x}{x^2 + 5x}$	◀ KFC
STEP 2	$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)}$	◀ factorise all four
STEP 3	$\frac{4 \times 10x}{5x}$	◀ cancel $(x-6)$, $(x+6)$, $(x+5)$
STEP 4	$= 8$	◀ multiply

ANSWER $p = 8$ ✓ CALC $x = 2$: $\frac{-112}{-160} \div \frac{14}{160} = 8$ **WATCH OUT**

× flipping the first fraction

✓ Keep the first, flip the second

Only the fraction after $\div$ turns upside down.**EXAM TIP**“ p is an integer” tells you every bracket must cancel — if one does not, a factorisation is wrong.**Check yourself**

calculator first · answers on page 19

2.1 Practice · MediumSimplify $\frac{25a}{8} \times \frac{2a}{5}$

Answer:

2.2 Practice · Very hardSimplify $\frac{3x+6}{x^2-3x-10} \div \frac{x+5}{x^3-25x}$

Answer:

2.3 Dr EslamSimplify $\frac{2x+6}{x^2-1} \div \frac{x+3}{x-1}$

Answer:

V3

VARIANT 3

**Add & Subtract:
Simple Bottoms**

Bottoms like 4, 6x, 2x: find the LCM, change each top, add.

SEEN IN PAST PAPERS

4 / 18

PRACTICE QUESTIONS: 7

 Your notes

Came up in: Jan 2022 P1HR Q12(b) • Nov 2023 P1H Q11 • May 2024 P1H Q12(b) • Jun 2025 P1H Q12

? What is it?**IN PLAIN WORDS**

To add or subtract, the fractions need the **same bottom**. Find the **LCM** of the bottoms (with letters: the LCM of 4 and 6x is 12x). Multiply each top by what its bottom was multiplied by. Then add the tops.

IT LOOKS LIKE

$$\frac{3}{4} + \frac{5-x}{6x} \quad \frac{2x+1}{4x} + \frac{7-5x}{3x} \quad \frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

LCM of 4 and 6x = 12x

$$\frac{3}{4} + \frac{5-x}{6x} \longrightarrow \frac{3 \times 3x}{12x} + \frac{(5-x) \times 2}{12x} = \frac{7x+10}{12x}$$

$4 \times 3x \qquad 6x \times 2$

Each bottom becomes 12x; its top is multiplied by the same thing.

How do I do it?**1 LCM**

The smallest thing every bottom goes into: 4 and 6x → 12x.

2 Change each top

Multiply each top by what its bottom was multiplied by.

3 One fraction

Add/subtract the tops (bracket them!) over the LCM.

4 Simplify

Expand, collect; cancel if the top factorises.

LCM FOR + AND - Never add the bottoms.

V3 Add & Subtract: Simple Bottoms

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Express ... as a single fraction in its simplest form”

→ **LCM bottom, change the tops, add**

Bottoms 4 and 6x

→ **LCM = 12x (not 24x)**

A minus before a fraction

→ **Bracket its top: $-(3x - 5)$** **WORKED EXAMPLE** **PAST PAPER**

November 2023 · Paper 1H · Q11 · 3 marks

Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest form.

STEP 1 LCM of 4 and 6x = 12x ◀ find the LCM

STEP 2 $\frac{3 \times 3x}{12x} + \frac{2(5-x)}{12x}$ ◀ change each top

STEP 3 $\frac{9x + 10 - 2x}{12x}$ ◀ one fraction

STEP 4 $\frac{7x + 10}{12x}$ ◀ collect

ANSWER $\frac{7x + 10}{12x}$

✓ CALC x = 2: both sides 1

WATCH OUT

$$\times \frac{3}{4} + \frac{5-x}{6x} = \frac{8-x}{4+6x}$$

$$\checkmark \frac{7x+10}{12x}$$

Never add tops and bottoms. Make the bottoms the same first.

EXAM TIP

4 of the 18 past-paper parts — always 3 marks: M1 for the common bottom, M1 for the tops, A1.

Check yourself

calculator first · answers on page 19

3.1 Jun 2025 P1H

Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction

Answer:

3.2 May 2024 P1H

Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction

Answer:

3.3 Jan 2022 P1HR

Express $\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$ as a single fraction

Answer:

V4

VARIANT 4

**Add & Subtract:
Bracket Bottoms**

Bottoms like $(x - 2)$ and $(x + 1)$: the common bottom is their product.

SEEN IN PAST PAPERS

0 / 18

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 12


Your notes
? What is it?**In plain words**

Two different brackets on the bottom: the common bottom is $(x - 2)(x + 1)$. Each top is multiplied by the **other** bottom — the **butterfly**.

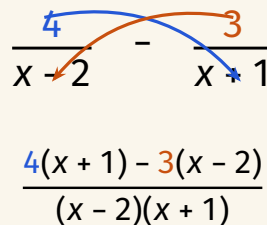
If a bottom factorises ($x^2 - 1$), factorise it first — the LCM may be smaller.

IT LOOKS LIKE

$$\frac{4}{x-2} - \frac{3}{x+1}$$

$$\frac{x+9}{x^2-1} + \frac{4}{x+1}$$

THE BUTTERFLY



$$\frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

How do I do it?**1 Factorise bottoms**

If a bottom factorises, do it — look for a shared bracket.

2 Common bottom

The product of the different brackets (each once).

3 Butterfly tops

Each top $\times$ the bracket it is missing. Minus in front: bracket it.

4 Simplify

Expand and collect the top; cancel if possible. Keep the bottom factorised.

THE BUTTERFLY Top $\times$ the other bottom.

V4 Add & Subtract: Bracket Bottoms

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

$$\frac{4}{x-2} - \frac{3}{x+1}$$

→ **Common bottom** $(x-2)(x+1)$, **butterfly**A bottom like $x^2 - 1$ next to $x + 1$ → **Factorise:** $(x+1)(x-1)$ — **LCM is** $(x+1)(x-1)$ “can be written in the form $\frac{a}{x-1}$ ”→ **The top factorises and a bracket cancels****WORKED EXAMPLE****PRACTICE**

Algebraic Fractions • Medium Q7 • 3 marks

Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction in its simplest form.**STEP 2** common bottom $(x-2)(x+1)$

◀ different brackets: multiply

$$\frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

◀ butterfly; bracket after the minus

$$\frac{4x+4-3x+6}{(x-2)(x+1)} = \frac{x+10}{(x-2)(x+1)}$$

◀ expand and collect

ANSWER
$$\frac{x+10}{(x-2)(x+1)}$$

✓ CALC $x = 3$: both sides $\frac{13}{4}$ **WATCH OUT**

$$\times 4(x+1) - 3(x-2) = x-2$$

$$\checkmark 4x+4-3x+6 = x+10$$

 $-3 \times -2 = +6$: the minus multiplies both terms of $(x-2)$.
**EXAM TIP**

Leave the bottom in brackets — it is “simplest form” and it shows the examiner your common bottom.

**Check yourself**

calculator first • answers on page 19

4.1 Practice • MediumWrite $\frac{5}{x-3} - \frac{4}{x+3}$ as one fraction

Answer:

4.2 Practice • MediumWrite $\frac{3}{x-1} + \frac{4}{x+2}$ as one fraction

Answer:

4.3 Practice • HardShow that $\frac{x+9}{x^2-1} + \frac{4}{x+1}$ can be written as $\frac{a}{x-1}$

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM**Mixed Operations**Several operations in one question:
BIDMAS order, then V1–V4.

SEEN IN PAST PAPERS

7 / 18

PRACTICE QUESTIONS: 5


Your notes

Came up in: Jan 2020 P2H Q24 • Oct/Nov 2020 P1H Q21 • Jan 2021 P2H Q21 • Jun 2022 P2HR Q23 • May 2023 P1HR Q23 • Jun 2024 P2HR Q26 • Nov 2024 P2H Q23

? What is it?**IN PLAIN WORDS**The exam's favourite: $a \div$ or $\times$ **and** $a -$ or $+$ in one question. Follow **BIDMAS**: do the bracket and the $\times \div$ part first (KFC, factorise, cancel) until it is **one simple fraction**; do the $+/-$ **last** with the LCM.**IT LOOKS LIKE**

$$4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right] \quad \frac{1}{3x - 2} \times \frac{9x^2 - 4}{3x^2 - 13x - 10} - \frac{7}{x - 1}$$

1 BRACKET: KFC

$$4 - \frac{(3x - 5)(4x - 1)}{3x^2 + x - 10}$$

2 FACTORISE, CANCEL

$$4 - \frac{4x - 1}{x + 2}$$

3 - LAST: LCM

$$\frac{4(x + 2) - (4x - 1)}{x + 2}$$

$$\frac{9}{x + 2}$$

Three stages, in BIDMAS order. Each stage is a method mark.

How do I do it?**1 Bracket first**Do the [] or the $\times \div$ part: KFC if there is a $\div$.**2 Factorise + cancel**

Factorise every top and bottom, cancel — one small fraction left.

3 + - last

Common bottom (LCM), change the tops (whole numbers get the bottom too).

4 Simplify

Expand, collect; give the form asked for.

BIDMAS: $\times \div$ FIRST, $+ -$ LAST One clean fraction before you add.

V5 Mixed Operations

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO $“4 - [\dots \div \dots]”$ → **Bracket first (KFC, cancel), subtract last** $“Show that \dots = n \text{ where } n \text{ is an integer}”$ → **All letters must cancel at the end****WORKED EXAMPLE** **PAST PAPER**

June 2024 · Paper 2HR · Q26 · 4 marks

Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

STEP 1+2 $(3x - 5) \times \frac{4x - 1}{(3x - 5)(x + 2)} = \frac{4x - 1}{x + 2}$

◀ bracket first: KFC, factorise, cancel

STEP 3+4 $\frac{4(x + 2) - (4x - 1)}{x + 2} = \frac{4x + 8 - 4x + 1}{x + 2}$

◀ subtract last: common bottom, collect

ANSWER $\frac{9}{x + 2}$

✓ CALC $x = 2$: both sides $\frac{9}{4}$ **WATCH OUT**

$$\times 4 - \frac{4x - 1}{x + 2} = \frac{4 - 4x - 1}{x + 2}$$

$$\checkmark \frac{4(x + 2) - (4x - 1)}{x + 2}$$

The 4 needs the bottom too, and the top after the minus needs a bracket.

**EXAM TIP**

7 of the 18 past-paper parts (4–5 marks).
Write each stage on its own line —
method marks come stage by stage.

**Check yourself**

calculator first · answers on page 19

5.1

Oct/Nov 2020 P1H

Express $\frac{1}{3x - 2} \times \frac{9x^2 - 4}{3x^2 - 13x - 10} - \frac{7}{x - 1}$
as one fraction

Answer:

5.2

Jan 2021 P2H

Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$ as
one fraction

Answer:

5.3

Nov 2024 P2H

Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$.
Find n .

Answer:

V6

VARIANT 6

Fraction in a Fraction

a depends on x , x depends on y :
 substitute, then clear the small fraction.

SEEN IN PAST PAPERS

4 / 18

PRACTICE QUESTIONS: 1


 Your notes

Came up in: May 2021 P1H Q21 • Jun 2022 P2H Q19 • Jun 2024 P2H Q20 • Nov 2025 P1H Q23

? What is it?**In plain words**

Two formulas: a in terms of x , and x in terms of y . **Substitute** the second into the first — now there is a fraction inside a fraction.

Clear it: multiply the big top and the big bottom by the small bottom. Then simplify.

IT LOOKS LIKE

$$a = \frac{14}{3x - 7}$$

$$x = \frac{7}{4y - 3}$$

$$a = \frac{14}{3\left(\frac{7}{4y - 3}\right) - 7}$$

↓ $\times(4y - 3)$ TOP AND BOTTOM

$$a = \frac{14(4y - 3)}{21 - 7(4y - 3)}$$

How do I do it?**1 Substitute**

Replace x by its fraction — in a bracket.

2 Clear

Multiply the big top and the big bottom by the small bottom.

3 Simplify

Expand the bottom, collect, factorise and cancel a common number.

4 Match the form

e.g. $\frac{py+q}{ry+s}$: read off p, q, r, s .

SUBSTITUTE, THEN CLEAR

Multiply top and bottom by the small bottom.

V6 Fraction in a Fraction

VARIANT 6 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Given that $a = \dots$ and $x = \dots$, express a in terms of y "

→ **Substitute, clear the small fraction**

"in the form $\frac{py+q}{ry+s}$ where ... integers"

→ **Simplify fully, then read the letters****WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 2H · Q19 · 3 marks

$a = \frac{14}{3x-7}$, $x = \frac{7}{4y-3}$. Express a in the form $\frac{py+q}{ry+s}$ where p, q, r, s are integers.

STEP 1 $a = \frac{14}{3 \cdot \frac{7}{4y-3} - 7}$

◀ substitute

STEP 2 $a = \frac{14(4y-3)}{21-7(4y-3)}$

◀ $\times(4y-3)$ top and bottom

STEP 3+4 $a = \frac{14(4y-3)}{14(3-2y)} = \frac{4y-3}{3-2y}$

◀ expand, factorise 14, cancel

ANSWER $a = \frac{4y-3}{3-2y}$

✓ $y = 1$: $x = 7$, $a = 1$ ✓**WATCH OUT**

✗ multiplying only the top by $(4y-3)$

✓ top and bottom $\times(4y-3)$

Both, so the value stays the same.

**EXAM TIP**

Check with a number: choose $y = 1$, find x , then a from both your answer and the original — they must agree.

**Check yourself**

calculator first · answers on page 19

6.1

May 2021 P1H

$x = \frac{5}{9y+5}$, $y = \frac{5}{5a-2}$.
Find x in terms of a .

Answer:

6.2

Nov 2025 P1H

$a = \frac{2x+5}{1-x}$, $x = \frac{5-2y}{3y}$.
Find a in terms of y .

Answer:

6.3

Jun 2024 P2H

$k = x - y$, $x = \frac{1}{4y}$. Write
 $\frac{5k}{x+2}$ in terms of y .

Answer:

V7

VARIANT 7

Solving with Fractions

An equation with fractions: multiply every term by the LCM — the fractions vanish.

SEEN IN PAST PAPERS

2 / 18

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2021 P2HR Q18 • Nov 2023 P2H Q24

? What is it?

In plain words

Now there is an =. Factorise the bottoms, find the LCM, and **multiply every term** — the right side too — by it. The fractions disappear.

Often a **quadratic** is left: ERCS (expand, rearrange to = 0), then solve. **Reject** any x that makes a bottom 0.

IT LOOKS LIKE

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

$$\frac{x}{x+4} + \frac{7}{x-2} = 1$$

$$\frac{5}{x+2} + \frac{3}{x(x+2)} = 2$$

* x(x+2) EVERY TERM

$\frac{5}{x+2}$
↓
5x

$+$

$\frac{3}{x(x+2)}$
↓
3

$=$

2
↓
2x(x+2)

How do I do it?

1 Factorise bottoms

$x^2 + 2x = x(x+2)$. LCM = $x(x+2)$.

2 * every term

Every term, both sides. Each bottom cancels.

3 ERCS

Expand, rearrange to = 0, then factorise (CC Rule) or the formula.

4 Check

Reject any answer that makes a bottom 0; check the rest with CALC.

* **EVERY TERM BY THE LCM** Then it is Topic 9 or Topic 10.

V7 Solving with Fractions

VARIANT 7 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Solve” + fractions, “show clear algebraic working”

→ × every term by the LCM

“Give your solutions to 3 s.f.”

→ The quadratic does not factorise: use the formula

A speed, length or time

→ Reject the negative or impossible answer

WORKED EXAMPLE PAST PAPER

January 2021 · Paper 2HR · Q18 · 5 marks

Solve $\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$. Show clear algebraic working.

STEP 1 $x^2 + 2x = x(x+2)$: LCM $x(x+2)$ ◀ factorise the bottoms

STEP 2 $5x + 3 = 2x(x+2)$ ◀ × $x(x+2)$ every term

STEP 3 $2x^2 - x - 3 = 0 \Rightarrow (2x-3)(x+1) = 0$ ◀ ERCS, CC Rule

STEP 4 $x = \frac{3}{2}$ or $x = -1$ ◀ neither makes a bottom 0

ANSWER $x = \frac{3}{2}$ or $x = -1$

✓ CALC $x = 1.5$: $\frac{10}{7} + \frac{4}{7} = 2$

WATCH OUT

× $5x + 3 = 2$ (the right side not multiplied)

✓ $5x + 3 = 2x(x+2)$

Every term means every term — the 2 on the right is multiplied by $x(x+2)$ too.

EXAM TIP

$x = 0$ or $x = -2$ would make a bottom zero: if you get one of them, reject it and say why.

Check yourself

calculator first · answers on page 19

7.1 Nov 2023 P2H

Solve $\frac{45x^3 - 80x}{3x^2 + x - 4} \times$
 $\left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$

Answer:

7.2 Practice · Hard

Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Answer:

7.3 Practice · Very hard

Solve $\frac{x}{x+4} + \frac{7}{x-2} = 1$

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

$$1.1 \quad \frac{4}{x-3}$$

$$1.2 \quad \frac{5x+4}{2x-3}$$

$$1.3 \quad 2x-2$$

$$2.1 \quad \frac{5a^2}{4}$$

$$2.2 \quad 3x$$

$$2.3 \quad \frac{2}{x+1}$$

$$3.1 \quad \frac{17x-6}{12x}$$

$$3.2 \quad \frac{31-14x}{12x}$$

$$3.3 \quad \frac{3x+16}{4x}$$

$$4.1 \quad \frac{x+27}{(x-3)(x+3)}$$

$$4.2 \quad \frac{7x+2}{(x-1)(x+2)}$$

$$4.3 \quad \frac{5}{x-1}$$

$$5.1 \quad \frac{34-6x}{(x-5)(x-1)}$$

$$5.2 \quad \frac{54}{5x+2}$$

$$5.3 \quad n = -19$$

$$6.1 \quad x = \frac{5a-2}{5a+7}$$

$$6.2 \quad a = \frac{10+11y}{5(y-1)}$$

$$6.3 \quad \frac{5-20y^2}{1+8y}$$

$$7.1 \quad x = \frac{7}{4}$$

$$7.2 \quad x = 3$$

$$7.3 \quad x = -12$$

THE WHOLE TOPIC ON ONE PAGE

Algebraic Fractions cheat sheet

**V1 Factorise & cancel**

$$\frac{(x+3)(x-3)}{(2x-1)(x+3)} = \frac{x-3}{2x-1}$$

Golden rule: factorise everything first.

V2 $\times \div$ KFC 1/18

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Keep · Flip · Change, then cancel.

V3 Simple bottoms 4/18

$$\frac{3}{4} + \frac{5-x}{6x} = \frac{7x+10}{12x}$$

LCM; change each top.

V4 Bracket bottoms

$$\frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

Butterfly: top $\times$ the other bottom.**V5 Mixed 7/18**

$$4 - \frac{4x-1}{x+2} = \frac{9}{x+2}$$

 $\times \div$ first (KFC, cancel), $+$ $-$ last.**V6 Fraction in a fraction 4/18** $\times(4y-3)$ top and bottom

Substitute, clear, simplify.

V7 Solving 2/18

$$5x+3 = 2x(x+2)$$

 $\times$ every term by the LCM; reject bottom = 0.**TOP 3 LOST MARKS**

1. cancelling terms: $\frac{x+4}{x+6} = \frac{4}{6} \times$
2. adding the bottoms $\times$
3. $4 - \frac{4x-1}{x+2} = \frac{4-4x-1}{x+2} \times$

BOTTOM LINE Factorise $\rightarrow$ cancel $\rightarrow$ KFC for $\div$ $\rightarrow$ LCM for $\pm$ $\rightarrow$ to solve, $\times$ every term by the LCM.